Task 1: Create an Image from a Running Container

1. Objective

To create a custom Docker image from a running container using the `docker commit` command and verify that the image retains the container's filesystem changes.

2. Steps Used

Step 1: Verify Docker installation

```
sudo docker --version
```

Step 2: Check running containers

```
sudo docker ps
```

Step 3: Create an Ubuntu container

```
sudo docker run -dit --name docker03-container ubuntu:latest bash
```

Step 4: Customize the container

Created a custom file inside the running container.

```
sudo docker exec docker03-container bash -c 'echo "Welcome to Docker by Asifshaik" > /docker03.txt'
```

Step 5: Verify the custom file

```
sudo docker exec docker03-container cat /docker03.txt
```

Output:

```
Welcome to Docker by Asifshaik
```

Step 6: Create a Docker image from the container

```
sudo docker commit docker03-container docker03-custom:1.0
```

Step 7: Verify the created image

```
sudo docker images
```

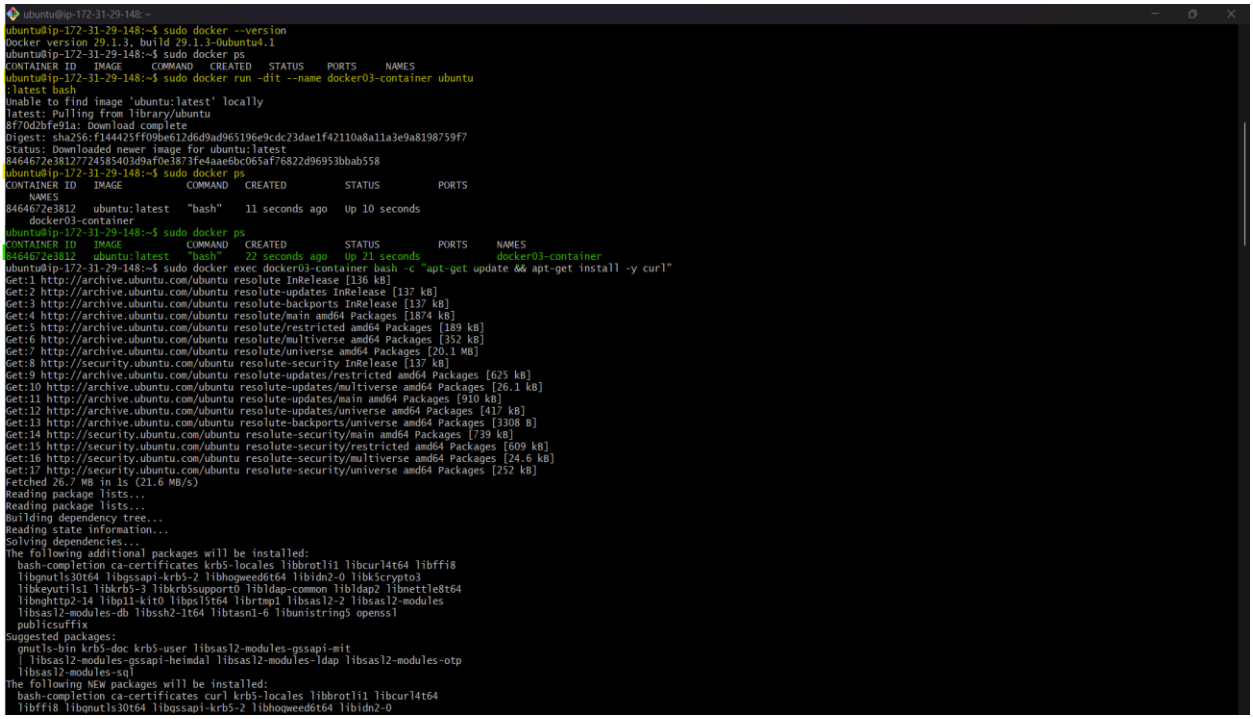
The image `docker03-custom:1.0` appeared in the image list.

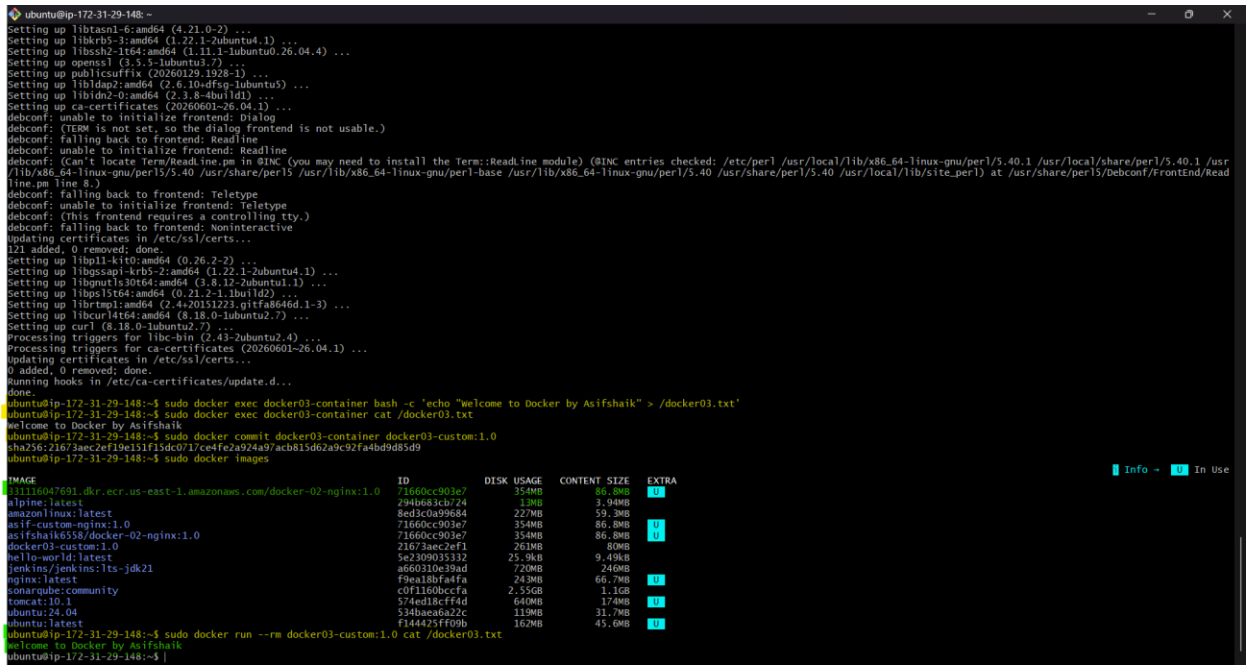
Step 8: Test the custom image

```
sudo docker run --rm docker03-custom:1.0 cat /docker03.txt
```

Output:

```
Welcome to Docker by Asifshaik
```





3. Observation

- The Ubuntu container was created and started successfully.
- The custom file was created inside the container.
- Docker generated the custom image `docker03-custom:1.0`.
- A new container created from that image successfully displayed the saved file content.

4. Conclusion

The task was completed successfully. The `docker commit` command was used to create a reusable Docker image from a modified running container. The image was tested successfully, confirming that the filesystem changes were preserved.

Task 2: Copy an Image from Your Local Machine to the Docker Server and Load It.

1. Objective

To export a Docker image from a local Windows machine, transfer it to a remote Ubuntu EC2 server using SCP, and load the image into the Docker image repository without using a container registry.

2. Steps Used

Step 1: Verify the local Docker image

On the local Windows machine using Git Bash, I checked the available Docker images.

```
docker images
```

Observation: The `hello-world:latest` image was available locally.

Step 2: Save the Docker image as a TAR file

I exported the image into a TAR archive.

```
docker save -o hello-world.tar hello-world:latest
```

To verify the archive:

```
ls -lh hello-world.tar
```

Observation: The `hello-world.tar` file was created successfully with a size of approximately 31 KB.

Step 3: Copy the image to the remote EC2 server

Using SCP, I transferred the archive from my local machine to the Ubuntu EC2 instance.

```
scp -i jenkins.pem ~/hello-world.tar \
ubuntu@ec2-54-225-22-113.compute-1.amazonaws.com:~
```

Observation: The TAR archive was transferred to the EC2 server.

Step 4: Verify the transferred file

On the EC2 terminal, I checked whether the archive was available.

```
ls -lh ~/hello-world.tar
```

Observation: The file was present at `/home/ubuntu/hello-world.tar`.

Step 5: Load the image into Docker

I imported the transferred archive into the Docker image repository.

```
sudo docker load -i ~/hello-world.tar
```

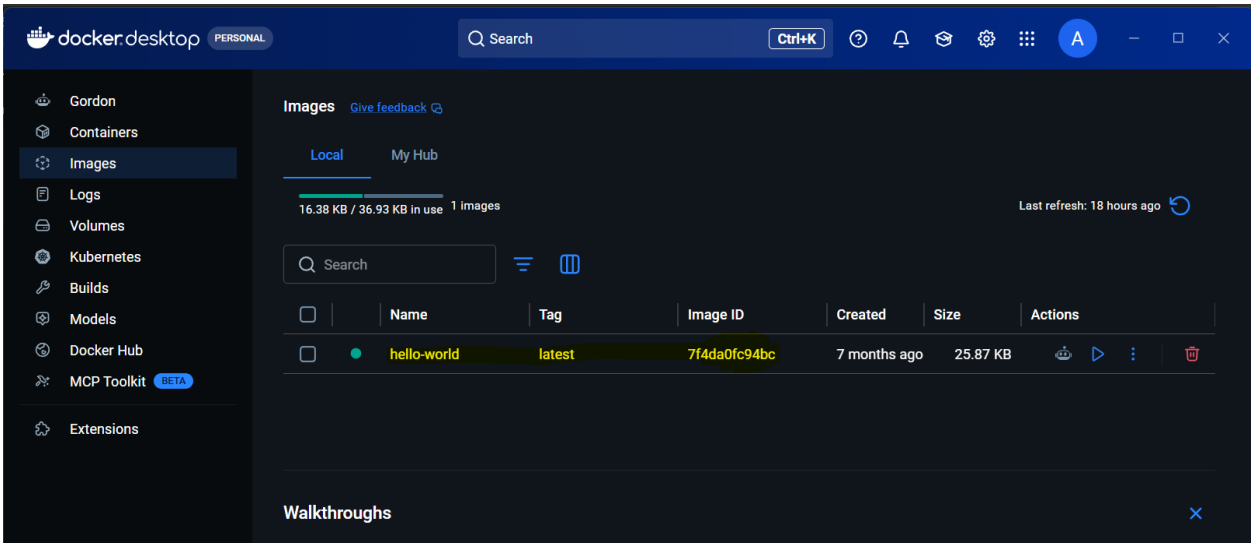
Observation: Docker displayed the message `Loaded image: hello-world:latest`.

Step 6: Verify the loaded image

I listed the Docker images on the EC2 server.

```
sudo docker images
```

Observation: The `hello-world:latest` image appeared in the image list, confirming successful import.



```
asifs@Asif_Shaik MINGW64 ~
$ docker images
IMAGE              ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest   7f4da0fc94bc       25.9kB         9.49kB   U

asifs@Asif_Shaik MINGW64 ~
$ docker save -o hello-world.tar hello-world:latest

asifs@Asif_Shaik MINGW64 ~
$ ls -lh hello-world.tar
-rw-r--r-- 1 asifs 197609 31K Oct 15 12:39 hello-world.tar

asifs@Asif_Shaik MINGW64 ~
$ scp -i jenkins.pem \
>   ~/hello-world.tar \
>   ubuntu@ec2-54-225-22-113.compute-1.amazonaws.com:~
Warning: Identity file jenkins.pem not accessible: No such file or directory.
ubuntu@ec2-54-225-22-113.compute-1.amazonaws.com: Permission denied (publickey).
scp: Connection closed

asifs@Asif_Shaik MINGW64 ~
$ cd ~/downloads

asifs@Asif_Shaik MINGW64 ~/downloads (master)
$ scp -i jenkins.pem \
>   ~/hello-world.tar \
>   ubuntu@ec2-54-225-22-113.compute-1.amazonaws.com:~
hello-world.tar                                                                    100%   31KB  54.9KB/s   00:00
```

```
ubuntu@ip-172-31-29-148:~$ ls -lh ~/hello-world.tar
-rw-r--r-- 1 ubuntu ubuntu 31K Oct 15 07:15 /home/ubuntu/hello-world.tar
ubuntu@ip-172-31-29-148:~$ sudo docker load -i ~/hello-world.tar
Loaded image: hello-world:latest
ubuntu@ip-172-31-29-148:~$ sudo docker images

IMAGE                                                                ID             DISK USAGE   CONTENT SIZE   EXTRA
331116047691.dkr.ecr.us-east-1.amazonaws.com/docker-02-nginx:1.0     71660cc903e7       354MB         86.8MB   U
alpine:latest                                                        294b683cb724        13MB         3.94MB
amazonlinux:latest                                                   8ed3c0a99684       227MB         59.3MB
asif-custom-nginx:1.0                                                71660cc903e7       354MB         86.8MB   U
asifshaik6558/docker-02-nginx:1.0                                    71660cc903e7       354MB         86.8MB   U
docker03-custom:1.0                                                  21673aec2ef1       261MB           80MB
hello-world:latest                                                   7f4da0fc94bc      25.9kB         9.49kB
jenkins/jenkins:lts-jdk21                                            a660310e39ad       720MB          246MB
nginx:latest                                                         f9ea18bfa4fa       243MB         66.7MB   U
sonarqube:community                                                  c0f1160bccfa      2.55GB          1.1GB
tomcat:10.1                                                          574ed18cff4d       640MB          174MB   U
ubuntu:24.04                                                         534baea6a22c       119MB         31.7MB
ubuntu:latest                                                        f144425ff09b       162MB         45.6MB   U
ubuntu@ip-172-31-29-148:~$
```

3. Overall Observation

- The Docker image was successfully exported from the local machine.
- The image archive was transferred to the remote EC2 server using SCP.
- The image was loaded successfully using `docker load`.
- The image became available on the remote Docker server without pulling it from Docker Hub or AWS ECR.

4. Conclusion

The Docker image transfer was completed successfully. I learned how to use `docker save` to export an image, SCP to transfer the image archive securely, and `docker load` to import the image on a remote Docker server. This method is useful when transferring Docker images between systems without using a container registry.

Task 3: Create a Docker Image Using Alpine and Customize It With Tomcat

1. Objective

To create a custom Docker image using Alpine Linux as the base image, install Java and Apache Tomcat, and run Tomcat inside a Docker container.

2. Steps Used

Step 1: Verify the Alpine image

```
sudo docker images alpine
```

Checked that the Alpine Linux image was available locally.

Step 2: Check the Alpine version

```
sudo docker run --rm alpine:latest cat /etc/alpine-release
```

Verified that the Alpine version was `3.24.2`.

Step 3: Create the project directory

```
mkdir -p ~/docker-03/alpine-tomcat
cd ~/docker-03/alpine-tomcat
```

Created a directory to store the Dockerfile.

Step 4: Create the Dockerfile

Created a Dockerfile containing these instructions:

- `FROM alpine:latest` — selects Alpine as the base image.
- `RUN apk add --no-cache openjdk21-jre curl tar` — installs Java and the required utilities.
- `ENV` — configures Java and Tomcat environment variables.
- `RUN curl ...` — downloads and extracts Apache Tomcat.
- `EXPOSE 8080` — documents the port used by Tomcat.
- `CMD ["catalina.sh", "run"]` — starts Tomcat in the foreground.

Step 5: Build the custom Docker image

```
sudo docker build -t alpine-tomcat:1.0 .
```

Built the custom image with the name `alpine-tomcat` and tag `1.0`.

Step 6: Run the container

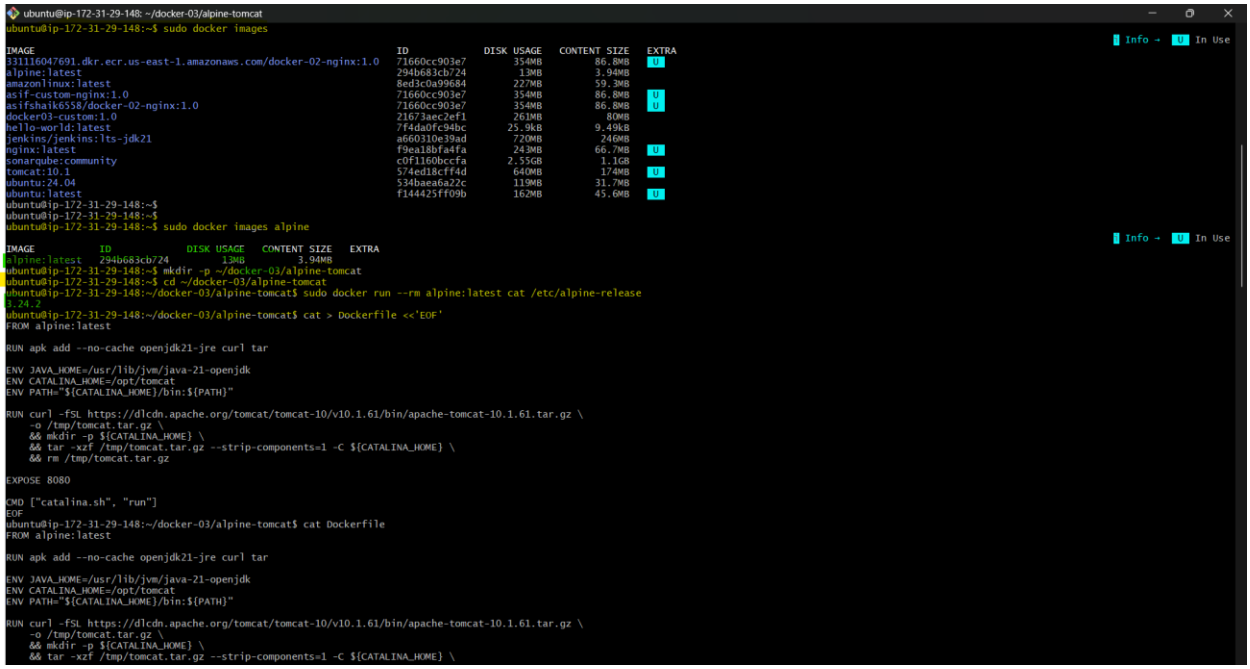
```
sudo docker run -d \
  --name alpine-tomcat-lab \
  -p 8084:8080 \
  alpine-tomcat:1.0
```

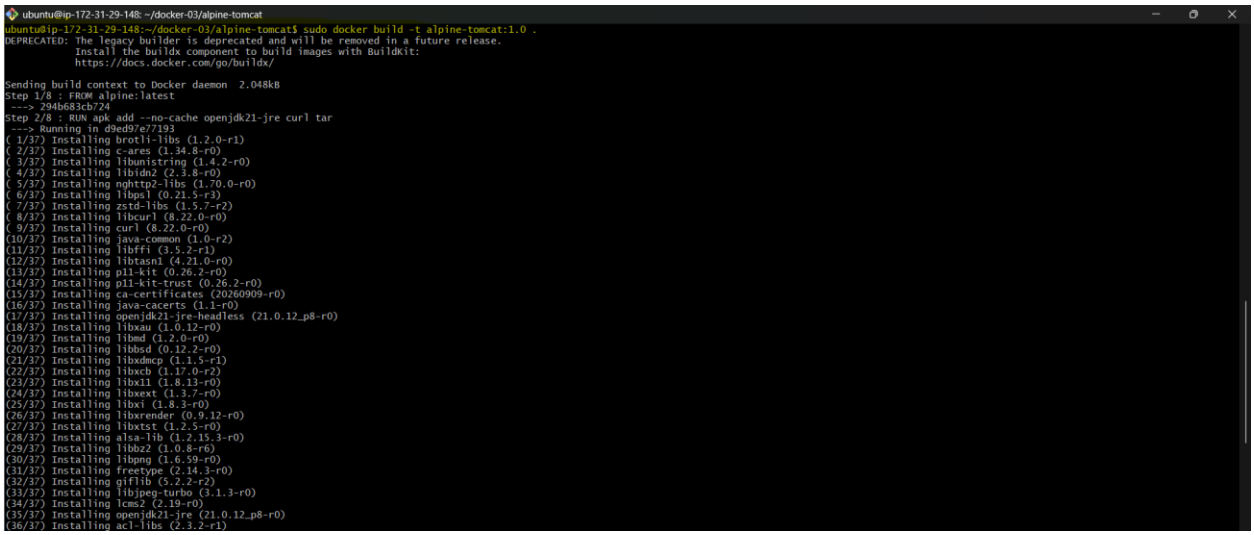
Started the container and mapped host port `8084` to container port `8080`.

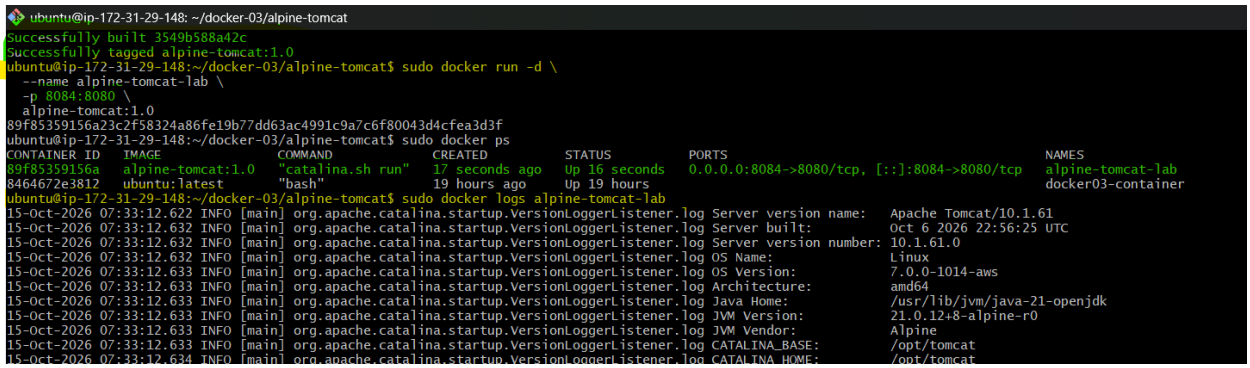
Step 7: Verify the container and test Tomcat

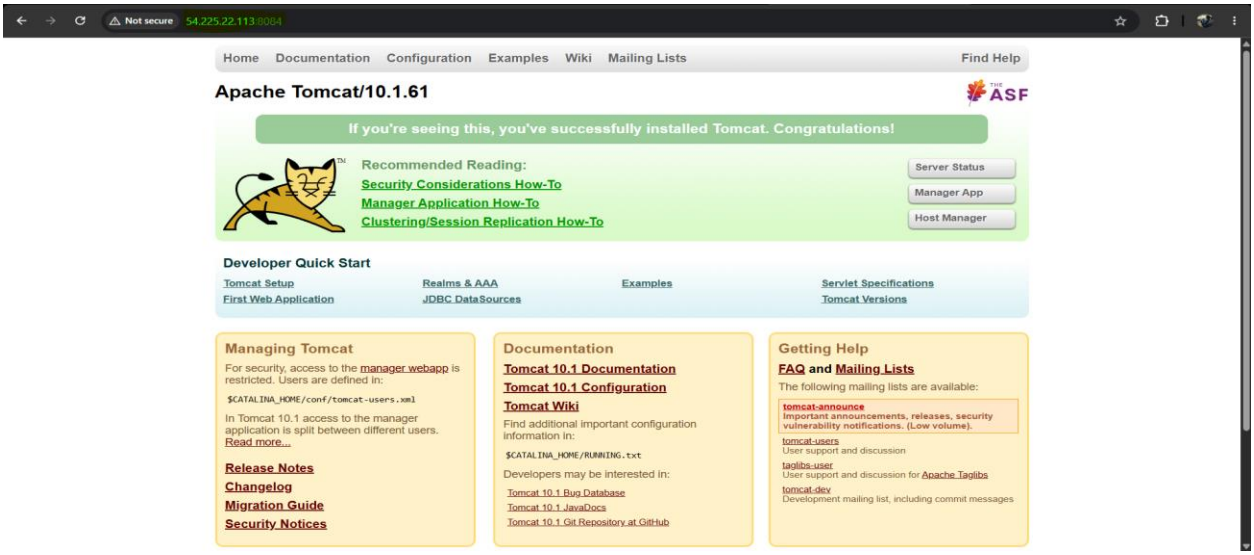
```
sudo docker ps
sudo docker logs alpine-tomcat-lab
curl -I http://localhost:8084
```

Checked the container status, Tomcat startup logs, and HTTP response.









3. Observation

- Alpine Linux was used as the lightweight base image.
- Java and Apache Tomcat were installed through the Dockerfile.

- The custom image `alpine-tomcat:1.0` was built successfully.
- The container was started with the port mapping `8084:8080`.
- The Tomcat application responded to the HTTP test.

4. Conclusion

Successfully created a custom Docker image using Alpine Linux and configured it to run Apache Tomcat. Learned how to install dependencies, configure environment variables, build an image using a Dockerfile, run a container, map ports, and verify the application using logs and an HTTP request.

Task 4: Create Single-Stage and Multi-Stage Docker Images

1. Objective

To create single-stage and multi-stage Docker images for a Java Spring Boot application using Dockerfiles, run the application in a container, and compare the image sizes to understand the benefits of multi-stage builds.

2. Steps Used

Step 1: Clone the GitHub repository

```
cd ~/docker-03
git clone https://github.com/betawins/multi-stage-example.git
cd multi-stage-example
```

Step 2: Create a backup of the original Dockerfile

```
cp Dockerfile Dockerfile.original
```

Step 3: Create the single-stage Dockerfile

Created `Dockerfile.single` with the following configuration:

```
FROM maven:3.8.8-eclipse-temurin-8

WORKDIR /app

COPY pom.xml .
```

```
COPY src ./src

RUN mvn clean package -DskipTests

EXPOSE 8080

ENTRYPOINT ["java", "-jar", "target/multi-stage-example-0.0.1-
SNAPSHOT.jar"]
```

Step 4: Build the single-stage image

```
sudo docker build -f Dockerfile.single -t multi-stage-single:v1 .
```

Step 5: Create the multi-stage Dockerfile

Created `Dockerfile.multi` using separate builder and runtime stages:

```
FROM maven:3.8.8-eclipse-temurin-8 AS builder

WORKDIR /app

COPY pom.xml .
COPY src ./src

RUN mvn clean package -DskipTests

FROM eclipse-temurin:8-jre

WORKDIR /app

COPY --from=builder /app/target/multi-stage-example-0.0.1-SNAPSHOT.jar
app.jar

EXPOSE 8080

ENTRYPOINT ["java", "-jar", "app.jar"]
```

Step 6: Build the multi-stage image

```
sudo docker build -f Dockerfile.multi -t multi-stage-multi:v1 .
```

Step 7: Compare the image sizes

```
sudo docker images multi-stage-single
sudo docker images multi-stage-multi
```

Step 8: Run the multi-stage container

```
sudo docker run -d --name multi-stage-lab -p 8086:8080 multi-stage-multi:v1
```

Step 9: Verify the container status

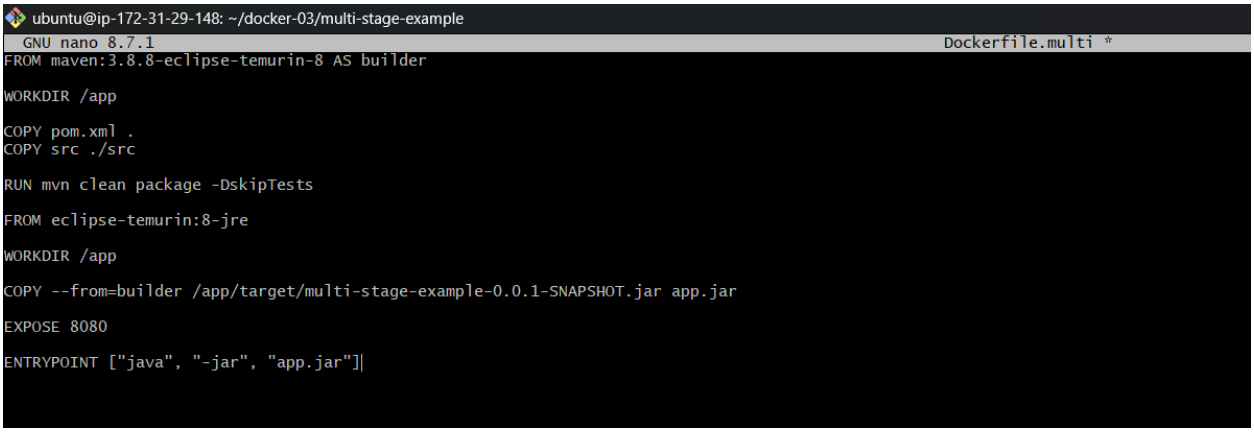
```
sudo docker ps
```

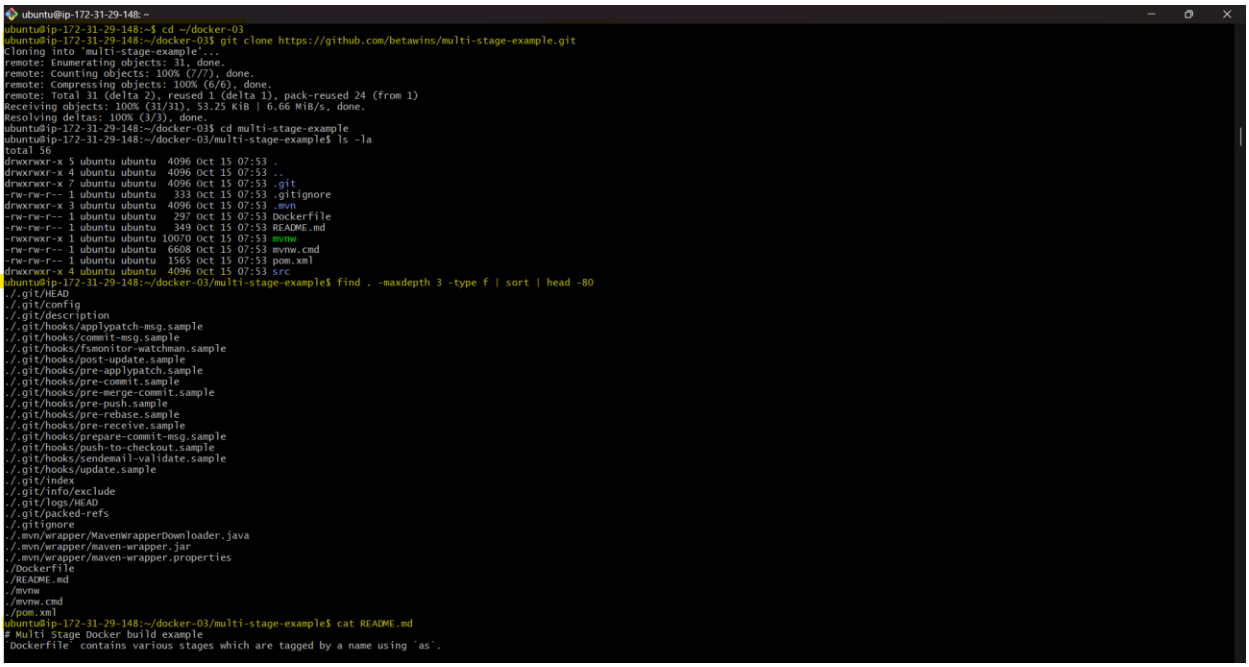
The container appeared with status `Up`, confirming that it was running.

Step 10: Check the application response and logs

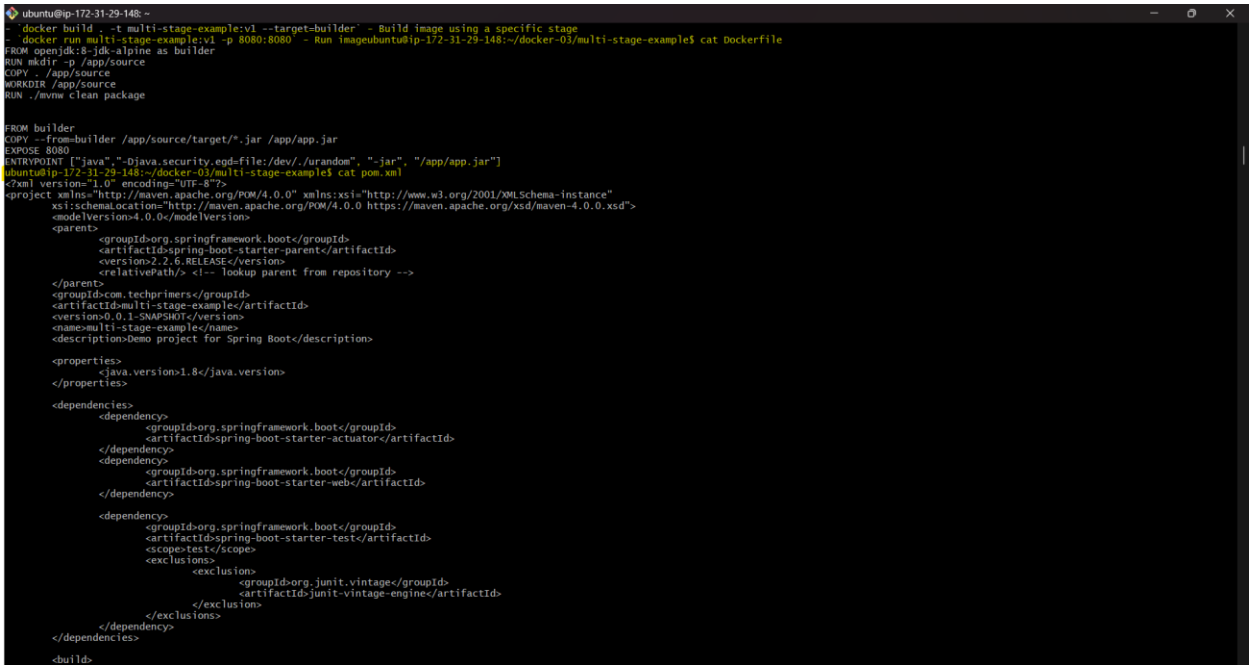
```
curl -I http://localhost:8086
sudo docker logs --tail 30 multi-stage-lab
```

The Spring Boot application started successfully. The root URL returned HTTP `404`, indicating that no handler was available for `/`; this did not mean the container had failed.





```
ubuntu@ip-172-31-29-148: ~
ubuntu@ip-172-31-29-148:~$ cd ~/docker-03
ubuntu@ip-172-31-29-148:~/docker-03$ git clone https://github.com/betawins/multi-stage-example.git
Cloning into 'multi-stage-example'...
remote: Enumerating objects: 31, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 31 (delta 2), reused 1 (delta 1), pack-reused 24 (from 1)
Receiving objects: 100% (31/31), 53.25 KiB | 6.66 MiB/s, done.
Resolving deltas: 100% (3/3), done.
ubuntu@ip-172-31-29-148:~/docker-03$ cd multi-stage-example
ubuntu@ip-172-31-29-148:~/docker-03/multi-stage-example$ ls -la
total 56
drwxrwxr-x 5 ubuntu ubuntu  4096 Oct 15 07:53 .
drwxrwxr-x 4 ubuntu ubuntu  4096 Oct 15 07:53 ..
drwxrwxr-x 7 ubuntu ubuntu  4096 Oct 15 07:53 .git
-rw-rw-r-- 1 ubuntu ubuntu   333 Oct 15 07:53 .gitignore
drwxrwxr-x 3 ubuntu ubuntu  4096 Oct 15 07:53 .mvn
-rw-rw-r-- 1 ubuntu ubuntu   297 Oct 15 07:53 Dockerfile
-rw-rw-r-- 1 ubuntu ubuntu   349 Oct 15 07:53 README.md
-rwxrwxr-x 1 ubuntu ubuntu 10070 Oct 15 07:53 mvnw
-rw-rw-r-- 1 ubuntu ubuntu  6608 Oct 15 07:53 mvnw.cmd
-rw-rw-r-- 1 ubuntu ubuntu  1565 Oct 15 07:53 pom.xml
drwxrwxr-x 4 ubuntu ubuntu  4096 Oct 15 07:53 src
ubuntu@ip-172-31-29-148:~/docker-03/multi-stage-example$ find . -maxdepth 3 -type f | sort | head -80
./.git/HEAD
./.git/config
./.git/description
./.git/hooks/applypatch-msg.sample
./.git/hooks/commit-msg.sample
./.git/hooks/fsmonitor-watchman.sample
./.git/hooks/post-update.sample
./.git/hooks/pre-applypatch.sample
./.git/hooks/pre-commit.sample
./.git/hooks/pre-merge-commit.sample
./.git/hooks/pre-push.sample
./.git/hooks/pre-rebase.sample
./.git/hooks/pre-receive.sample
./.git/hooks/prepare-commit-msg.sample
./.git/hooks/push-to-checkout.sample
./.git/hooks/sendemail-validate.sample
./.git/hooks/update.sample
./.git/index
./.git/info/exclude
./.git/logs/HEAD
./.git/packed-refs
./.gitignore
./.mvn/wrapper/MavenWrapperDownloader.java
./.mvn/wrapper/maven-wrapper.jar
./.mvn/wrapper/maven-wrapper.properties
./Dockerfile
./README.md
./mvnw
./mvnw.cmd
./pom.xml
ubuntu@ip-172-31-29-148:~/docker-03/multi-stage-example$ cat README.md
# Multi Stage Docker build example
`Dockerfile` contains various stages which are tagged by a name using `as`.
```



```
ubuntu@ip-172-31-29-148: ~
- `docker build . -t multi-stage-example:v1 --target=builder` - Build image using a specific stage
- `docker run multi-stage-example:v1 -p 8080:8080` - Run imageubuntu@ip-172-31-29-148:~/docker-03/multi-stage-example$ cat Dockerfile
FROM openjdk:8-jdk-alpine as builder
RUN mkdir -p /app/source
COPY . /app/source
WORKDIR /app/source
RUN ./mvnw clean package

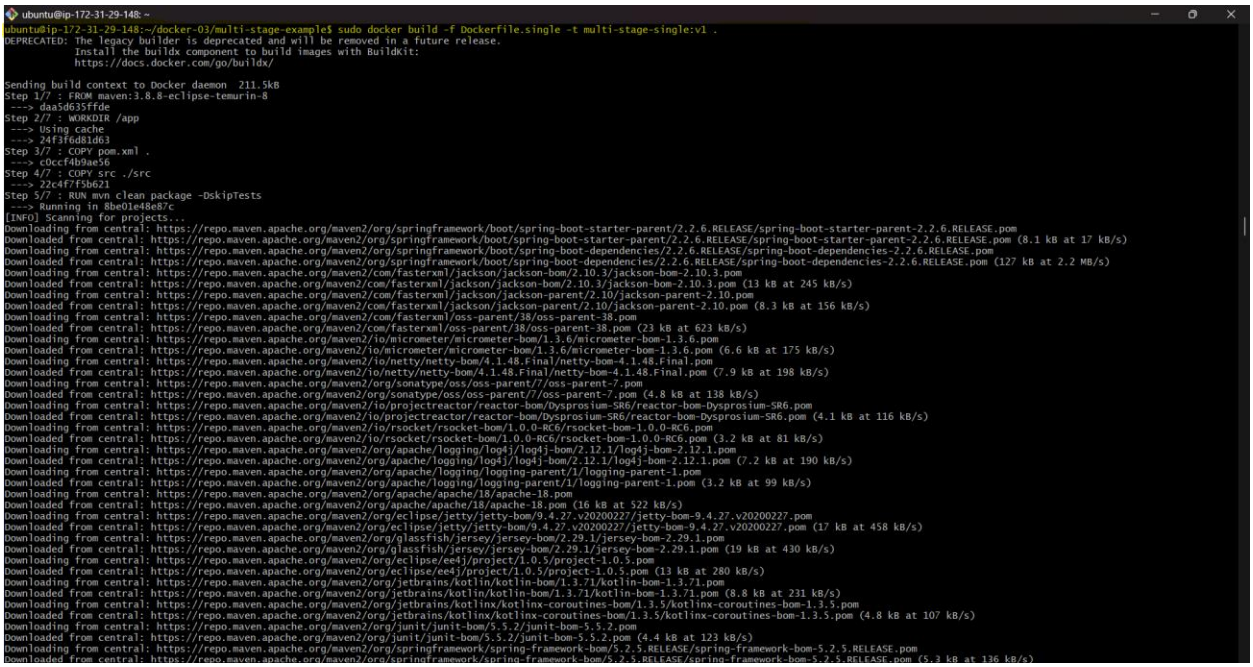

FROM builder
COPY --from=builder /app/source/target/*.jar /app/app.jar
EXPOSE 8080
ENTRYPOINT ["java","-Djava.security.egd=file:/dev/./urandom", "-jar", "/app/app.jar"]
ubuntu@ip-172-31-29-148:~/docker-03/multi-stage-example$ cat pom.xml
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-4.0.0.xsd">
        <modelVersion>4.0.0</modelVersion>
        <parent>
                <groupId>org.springframework.boot</groupId>
                <artifactId>spring-boot-starter-parent</artifactId>
                <version>2.2.6.RELEASE</version>
                <relativePath/> <!-- lookup parent from repository -->
        </parent>
        <groupId>com.techprimers</groupId>
        <artifactId>multi-stage-example</artifactId>
        <version>0.0.1-SNAPSHOT</version>
        <name>multi-stage-example</name>
        <description>Demo project for Spring Boot</description>

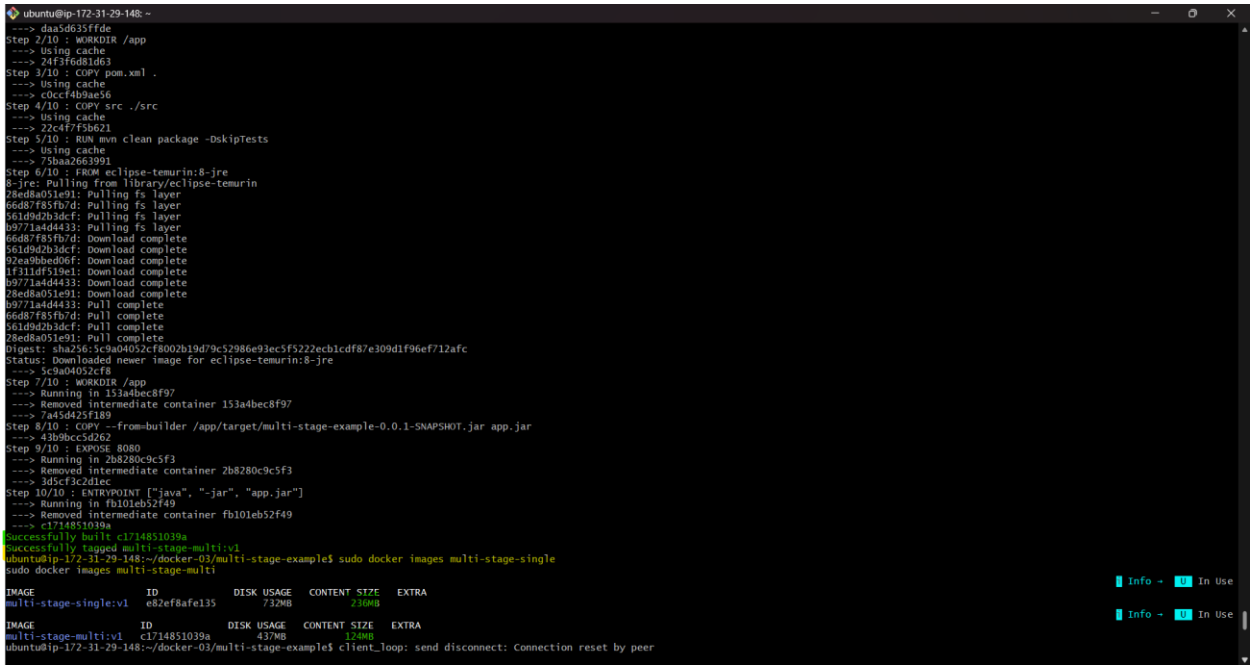
        <properties>
                <java.version>1.8</java.version>
        </properties>

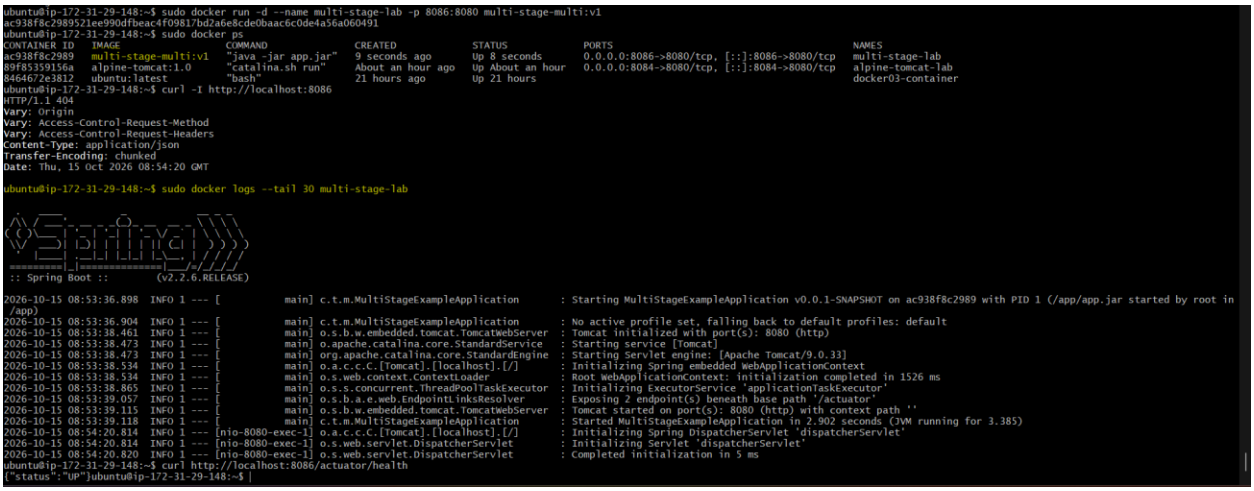
        <dependencies>
                <dependency>
                        <groupId>org.springframework.boot</groupId>
                        <artifactId>spring-boot-starter-actuator</artifactId>
                </dependency>
                <dependency>
                        <groupId>org.springframework.boot</groupId>
                        <artifactId>spring-boot-starter-web</artifactId>
                </dependency>

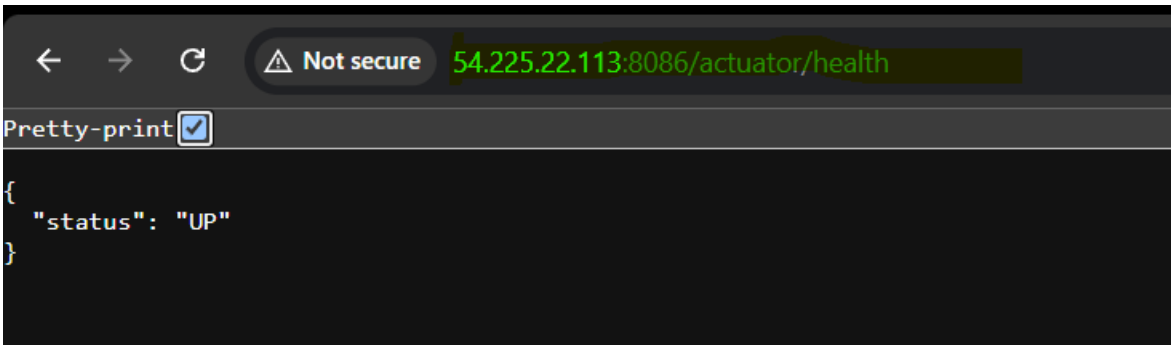
                <dependency>
                        <groupId>org.springframework.boot</groupId>
                        <artifactId>spring-boot-starter-test</artifactId>
                        <scope>test</scope>
                        <exclusions>
                                <exclusion>
                                        <groupId>org.junit.vintage</groupId>
                                        <artifactId>junit-vintage-engine</artifactId>
                                </exclusion>
                        </exclusions>
                </dependency>
        </dependencies>

        <build>
```


ubuntu@ip-172-31-29-148: ~
ubuntu@ip-172-31-29-148:~/docker-03/multi-stage-example$ sudo docker build -f Dockerfile.single -t multi-stage-single:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/
Sending build context to Docker daemon 211.5kB
Step 1/7 : FROM maven:3.8.8-eclipse-temurin-8
---> daa5d635ffde
Step 2/7 : WORKDIR /app
---> Using cache
---> 24f3f6d81d63
Step 3/7 : COPY pom.xml .
---> c0ccf4b9ae56
Step 4/7 : COPY src ./src
---> 22c4f7f5b621
Step 5/7 : RUN mvn clean package -DskipTests
---> Running in 8be01e48e87c
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-starter-parent/2.2.6.RELEASE/spring-boot-starter-parent-2.2.6.RELEASE.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-starter-parent/2.2.6.RELEASE/spring-boot-starter-parent-2.2.6.RELEASE.pom (8.1 kB at 17 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-dependencies/2.2.6.RELEASE/spring-boot-dependencies-2.2.6.RELEASE.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/boot/spring-boot-dependencies/2.2.6.RELEASE/spring-boot-dependencies-2.2.6.RELEASE.pom (127 kB at 2.2 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/com/fasterxml/jackson/jackson-bom/2.10.3/jackson-bom-2.10.3.pom
Downloaded from central: https://repo.maven.apache.org/maven2/com/fasterxml/jackson/jackson-bom/2.10.3/jackson-bom-2.10.3.pom (13 kB at 245 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/com/fasterxml/jackson/jackson-parent/2.10/jackson-parent-2.10.pom
Downloaded from central: https://repo.maven.apache.org/maven2/com/fasterxml/jackson/jackson-parent/2.10/jackson-parent-2.10.pom (8.3 kB at 156 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/com/fasterxml/oss-parent/38/oss-parent-38.pom
Downloaded from central: https://repo.maven.apache.org/maven2/com/fasterxml/oss-parent/38/oss-parent-38.pom (23 kB at 623 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/io/micrometer/micrometer-bom/1.3.6/micrometer-bom-1.3.6.pom
Downloaded from central: https://repo.maven.apache.org/maven2/io/micrometer/micrometer-bom/1.3.6/micrometer-bom-1.3.6.pom (6.6 kB at 175 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/io/netty/netty-bom/4.1.48.Final/netty-bom-4.1.48.Final.pom
Downloaded from central: https://repo.maven.apache.org/maven2/io/netty/netty-bom/4.1.48.Final/netty-bom-4.1.48.Final.pom (7.9 kB at 198 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/sonatype/oss/oss-parent/7/oss-parent-7.pom
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Downloading from central: https://repo.maven.apache.org/maven2/io/rsocket/rsocket-bom/1.0.0-RC6/rsocket-bom-1.0.0-RC6.pom
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Downloading from central: https://repo.maven.apache.org/maven2/org/apache/logging/log4j/log4j-bom/2.12.1/log4j-bom-2.12.1.pom
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Downloading from central: https://repo.maven.apache.org/maven2/org/apache/logging/logging-parent/1/logging-parent-1.pom
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Downloading from central: https://repo.maven.apache.org/maven2/org/apache/apache/18/apache-18.pom
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Downloading from central: https://repo.maven.apache.org/maven2/org/eclipse/jetty/jetty-bom/9.4.27.v20200227/jetty-bom-9.4.27.v20200227.pom
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Downloading from central: https://repo.maven.apache.org/maven2/org/glassfish/jersey/jersey-bom/2.29.1/jersey-bom-2.29.1.pom
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Downloading from central: https://repo.maven.apache.org/maven2/org/eclipse/ee4j/project/1.0.5/project-1.0.5.pom
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Downloading from central: https://repo.maven.apache.org/maven2/org/jetbrains/kotlin/kotlin-bom/1.3.71/kotlin-bom-1.3.71.pom
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Downloading from central: https://repo.maven.apache.org/maven2/org/jetbrains/kotlinx/kotlinx-coroutines-bom/1.3.5/kotlinx-coroutines-bom-1.3.5.pom
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Downloading from central: https://repo.maven.apache.org/maven2/org/junit/junit-bom/5.5.2/junit-bom-5.5.2.pom
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Downloading from central: https://repo.maven.apache.org/maven2/org/springframework/spring-framework-bom/5.2.5.RELEASE/spring-framework-bom-5.2.5.RELEASE.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/spring-framework-bom/5.2.5.RELEASE/spring-framework-bom-5.2.5.RELEASE.pom (5.3 kB at 136 kB/s)


ubuntu@ip-172-31-29-148: ~
---> daa5d635ffde
Step 2/10 : WORKDIR /app
---> Using cache
---> 24f3f6d81d63
Step 3/10 : COPY pom.xml .
---> Using cache
---> c0ccf4b9ae56
Step 4/10 : COPY src ./src
---> Using cache
---> 22c4f7f5b621
Step 5/10 : RUN mvn clean package -DskipTests
---> Using cache
---> 75baa2663991
Step 6/10 : FROM eclipse-temurin:8-jre
8-jre: Pulling from library/eclipse-temurin
28ed8a051e91: Pulling fs layer
66d87f85fb7d: Pulling fs layer
561d9d2b3dcf: Pulling fs layer
b9771a4d4433: Pulling fs layer
66d87f85fb7d: Download complete
561d9d2b3dcf: Download complete
92ea9bbed06f: Download complete
1f311df519e1: Download complete
b9771a4d4433: Download complete
28ed8a051e91: Download complete
b9771a4d4433: Pull complete
66d87f85fb7d: Pull complete
561d9d2b3dcf: Pull complete
28ed8a051e91: Pull complete
Digest: sha256:5c9a04052cf8002b19d79c52986e93ec5f5222ecb1cdf87e309d1f96ef712afc
Status: Downloaded newer image for eclipse-temurin:8-jre
---> 5c9a04052cf8
Step 7/10 : WORKDIR /app
---> Running in 153a4bec8f97
---> Removed intermediate container 153a4bec8f97
---> 7a45d425f189
Step 8/10 : COPY --from=builder /app/target/multi-stage-example-0.0.1-SNAPSHOT.jar app.jar
---> 43b9bcc5d262
Step 9/10 : EXPOSE 8080
---> Running in 2b8280c9c5f3
---> Removed intermediate container 2b8280c9c5f3
---> 3d5cf3c2d1ec
Step 10/10 : ENTRYPOINT ["java", "-jar", "app.jar"]
---> Running in fb101eb52f49
---> Removed intermediate container fb101eb52f49
---> c1714851039a
Successfully built c1714851039a
Successfully tagged multi-stage-multi:v1
ubuntu@ip-172-31-29-148:~/docker-03/multi-stage-example$ sudo docker images multi-stage-single
sudo docker images multi-stage-multi
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
multi-stage-single:v1 e82ef8afe135 732MB 230MB
Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
multi-stage-multi:v1 c1714851039a 437MB 124MB
Info → U In Use
ubuntu@ip-172-31-29-148:~/docker-03/multi-stage-example$ client_loop: send disconnect: Connection reset by peer





3. Observation

Image	Disk usage	Content size
Single-stage	732 MB	236 MB
Multi-stage	437 MB	124 MB
Difference	295 MB less	112 MB less

- The single-stage image includes the Maven build environment and application.
- The multi-stage build uses Maven to compile the application and copies only the generated JAR into a separate Java runtime image.
- The multi-stage image reduced disk usage by approximately 40% compared with the single-stage image.

- The multi-stage container started successfully, and the application logs confirmed that Spring Boot was listening on port `8080`.

4. Conclusion

Successfully created and compared single-stage and multi-stage Docker images for a Java Spring Boot application. The multi-stage build produced a smaller image by excluding build tools from the final runtime image. This demonstrates how multi-stage Docker builds can reduce image size and improve deployment efficiency.

Task 5: Install Docker Compose and Run a Sample Application

1. Objective

To install Docker Compose on an Ubuntu EC2 instance and use a Compose YAML file to deploy and verify a sample Nginx web application.

2. Steps Used

Step 1: Check Docker and system architecture

```
uname -m
sudo docker --version
```

Observed architecture: `x86_64` Docker version: `29.1.3`

Step 2: Install Docker Compose

Created the Docker CLI plugin directory:

```
mkdir -p ~/.docker/cli-plugins
```

Downloaded the Compose plugin:

`curl -SL` https://github.com/docker/compose/releases/latest/download/docker-compose-linux-x86_64 `-o ~/.docker/cli-plugins/docker-compose`

Granted execution permission:

```
chmod +x ~/.docker/cli-plugins/docker-compose
```

Verified the installation:

```
docker compose version
```

Output: `Docker Compose version v5.6.0`

Step 3: Create the application directory

```
mkdir -p ~/docker-03/compose-app
cd ~/docker-03/compose-app
```

Step 4: Create the Compose YAML file

Created `compose.yaml` with the following configuration:

```
services:
  web:
    image: nginx:latest
    ports:
      - "8087:80"
```

Step 5: Start the application

```
sudo /home/ubuntu/.docker/cli-plugins/docker-compose up -d
```

Step 6: Verify the container

```
sudo /home/ubuntu/.docker/cli-plugins/docker-compose ps
```

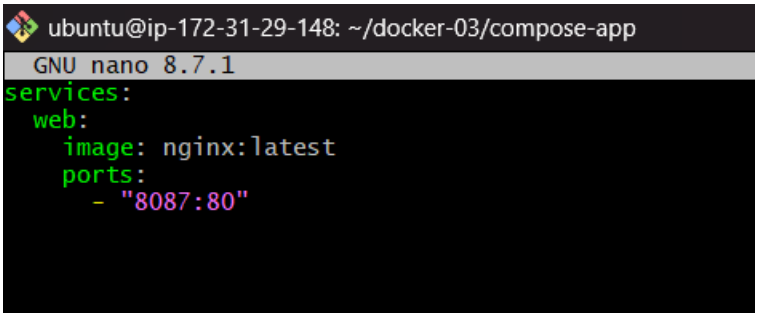
Observed container: `compose-app-web-1` Status: Up Port mapping: `8087:80`

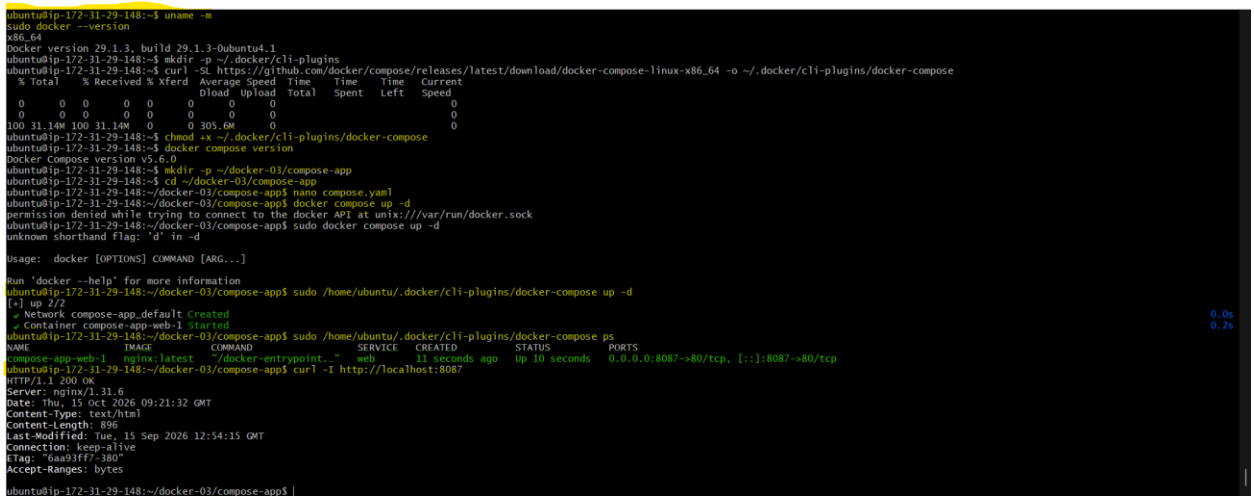
Step 7: Test the Nginx application

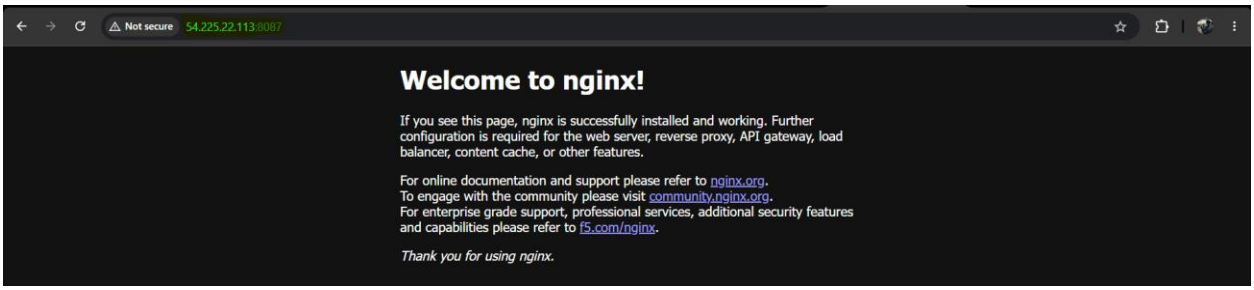
```
curl -I http://localhost:8087
```

Observed response:

```
HTTP/1.1 200 OK
Server: nginx/1.31.6
Content-Type: text/html
```







3. Observation

- Docker Compose was installed successfully.
- The `compose.yaml` file defined an Nginx service and its port mapping.
- Docker Compose created the application network and container.
- The container was running and accessible through host port `8087`.
- Nginx returned HTTP 200 OK, confirming successful access to the web server.

4. Conclusion

Docker Compose was successfully installed and used to deploy an Nginx application through a YAML configuration file. The container ran successfully, and the HTTP response confirmed that the application was working.

Docker Compose simplifies container management by allowing services, images, ports, and other configurations to be defined in a single file.

Task 6: Scan Docker Images Pushed to a Docker Registry Using Docker Scout

1. Objective

To scan a Docker image pushed to Docker Hub using Docker Scout and identify security vulnerabilities in the image's packages.

2. Steps Used

Step 1: Check the Docker image

The image used for scanning was the custom Nginx image pushed to Docker Hub in the previous task.

```
sudo docker images
```

Image name: `asifshaik6558/docker-02-nginx:1.0`

Step 2: Check Docker Scout

```
sudo docker scout version
```

Initially, Docker Scout was not available in the Docker CLI.

Step 3: Install Docker Scout

Downloaded the installation script:

```
curl -fsSL https://raw.githubusercontent.com/docker/scout-cli/main/install.sh -o install-scout.sh
```

Executed the script:

```
sh install-scout.sh
```

Docker Scout CLI was installed successfully.

Step 4: Verify the installation

Because Docker Scout was installed in the Ubuntu user's Docker configuration directory, used the following command with `sudo`:

```
sudo env DOCKER_CONFIG=/home/ubuntu/.docker docker scout version
```

Result: Docker Scout version `v1.26.0` was displayed.

Step 5: Log in to Docker Hub

```
docker login
```

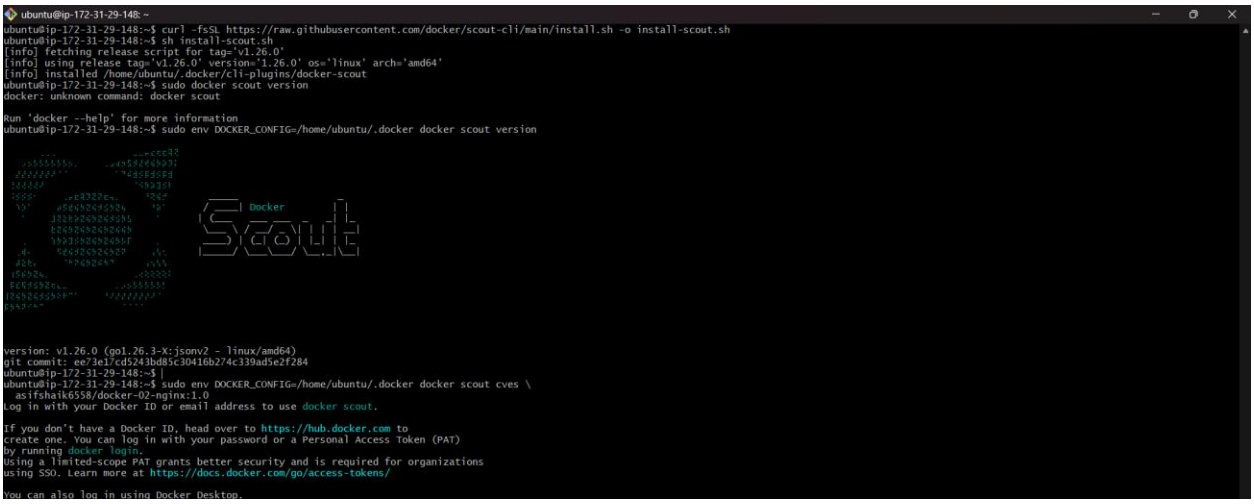
Completed the Docker Hub login process.

Step 6: Scan the Docker image

Executed the following command to scan the image for vulnerabilities:

```
sudo env DOCKER_CONFIG=/home/ubuntu/.docker docker scout cves
asifshaik6558/docker-02-nginx:1.0
```

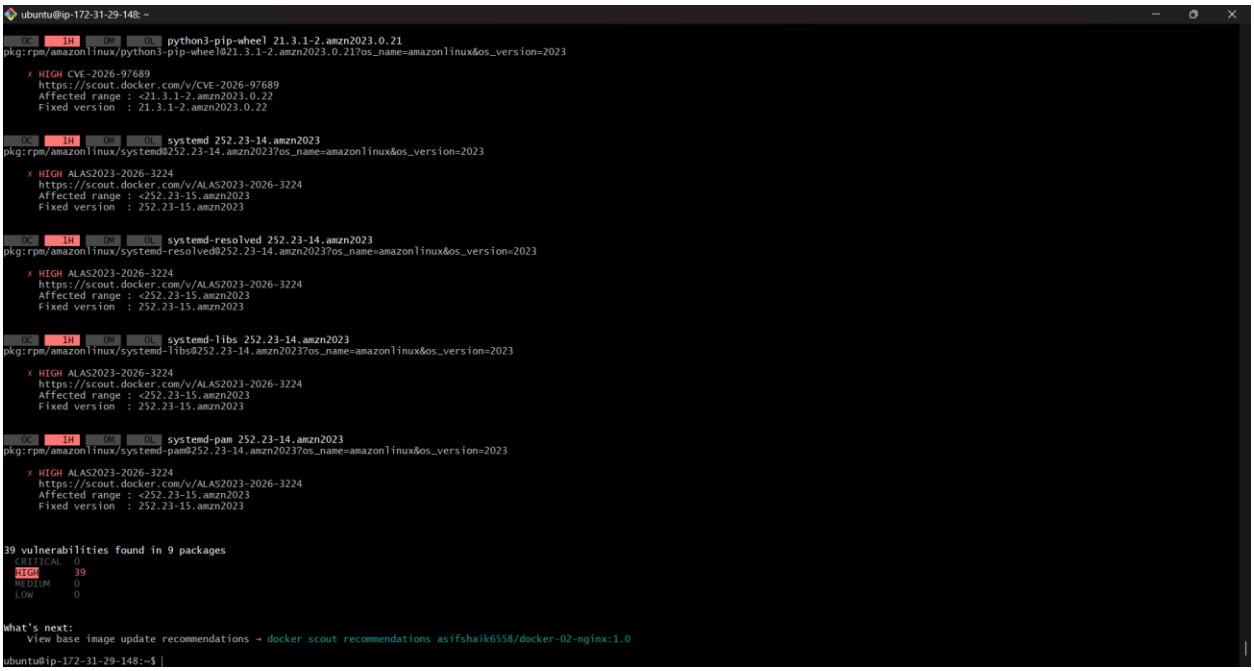
Docker Scout analyzed the image packages and displayed the detected security vulnerabilities, their severity, and available fixed versions.



```
ubuntu@ip-172-31-29-148: ~
ubuntu@ip-172-31-29-148:~$ curl -fsSL https://raw.githubusercontent.com/docker/scout-cli/main/install.sh -o install-scout.sh
ubuntu@ip-172-31-29-148:~$ sh install-scout.sh
[info] fetching release script for tag='v1.26.0'
[info] using release tag='v1.26.0' version='1.26.0' os='linux' arch='amd64'
[info] installed /home/ubuntu/.docker/cli-plugins/docker-scout
ubuntu@ip-172-31-29-148:~$ sudo docker scout version
docker: unknown command: docker scout

Run 'docker --help' for more information
ubuntu@ip-172-31-29-148:~$ sudo env DOCKER_CONFIG=/home/ubuntu/.docker docker scout version

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```



3. Observation

The Docker Scout scan produced the following results:

Parameter	Observation
Image	`asifshaik6558/docker-02-nginx:1.0`
Image size	Approximately 87 MB
Packages indexed	188
Affected packages	9
Reported vulnerability findings	39
Critical severity	0

High severity	39
Medium severity	0
Low severity	0

- Several findings were related to OpenSSL packages and systemd packages.
- Docker Scout displayed fixed versions for some affected packages.
- The 39 findings can include overlapping reports across related packages; they should not be interpreted as 39 distinct CVEs.
- No critical-severity findings were reported in this scan.

4. Conclusion

Docker Scout was successfully installed and used to scan the Docker image pushed to Docker Hub. The scan identified high-severity security findings in several packages. This task demonstrated how Docker Scout helps identify vulnerabilities in container images before deployment and provides information about available package fixes.

Task 7: Scan Docker Images Pushed to AWS ECR

1. Objective

To push a Docker image to Amazon Elastic Container Registry (ECR), scan the image for security vulnerabilities, and identify the severity levels of the detected vulnerabilities.

2. Steps Used

Step 1: Verify the existing Docker image

```
sudo docker images
```

Verified that the custom Nginx image `asif-custom-nginx:1.0` was available locally.

Step 2: Tag the image for AWS ECR

```
sudo docker tag asif-custom-nginx:1.0 331116047691.dkr.ecr.us-east-1.amazonaws.com/docker-02-nginx:1.0
```

Assigned the ECR repository URI to the existing image.

Step 3: Create and attach the IAM role

Created the IAM role `EC2-ECR-Push-Role` with EC2 as the trusted service and attached the `AmazonEC2ContainerRegistryPowerUser` policy.

Step 4: Verify AWS credentials

```
aws sts get-caller-identity
```

Confirmed that the EC2 instance was using the IAM role successfully.

Step 5: Log in to AWS ECR

```
aws ecr get-login-password --region us-east-1 | sudo env DOCKER_CONFIG=/home/ubuntu/.docker docker login --username AWS --password-stdin 331116047691.dkr.ecr.us-east-1.amazonaws.com
```

Authenticated Docker with AWS ECR.

Step 6: Push the image to ECR

```
sudo env DOCKER_CONFIG=/home/ubuntu/.docker docker push 331116047691.dkr.ecr.us-east-1.amazonaws.com/docker-02-nginx:1.0
```

Successfully pushed the image to the `docker-02-nginx` repository.

Step 7: Scan the image

1. Opened AWS Console and navigated to Amazon ECR.
2. Opened the `docker-02-nginx` repository.
3. Selected image tag `1.0`.
4. Started the image scan.
5. Waited for the scan status to show Complete.
6. Reviewed the vulnerability findings and severity levels.

```
ubuntu@ip-172-31-29-148:~$ sudo docker images

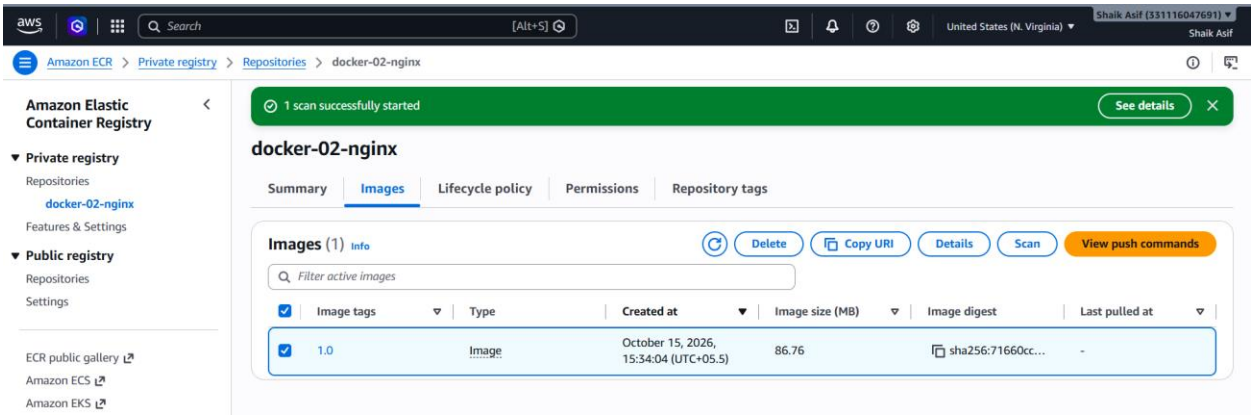
IMAGE                                                                 ID             DISK USAGE   CONTENT SIZE   EXTRA
331116047691.dkr.ecr.us-east-1.amazonaws.com/docker-02-nginx:1.0      71660cc903e7       354MB         86.8MB    U
alpine-tomcat:1.0                                                     3549b588a42c       341MB         92.2MB    U
alpine:latest                                                         294b683cb724        13MB         3.94MB
amazonlinux:latest                                                    8ed3c0a99684       227MB         59.3MB
asif-custom-nginx:1.0                                                 71660cc903e7       354MB         86.8MB    U
asifshaik6558/docker-02-nginx:1.0                                     71660cc903e7       354MB         86.8MB    U
docker03-custom:1.0                                                   21673aec2ef1       261MB           80MB
eclipse-temurin:8-jre                                                 5c9a04052cf8       405MB          112MB
hello-world:latest                                                    7f4da0fc94bc      25.9kB         9.49kB
jenkins/jenkins:lts-jdk21                                             a660310e39ad       720MB          246MB
maven:3.8.8-eclipse-temurin-8                                         daa5d635ffde       590MB          174MB
multi-stage-multi:v1                                                  c1714851039a       437MB          124MB    U
multi-stage-single:v1                                                 e82ef8afe135       732MB          236MB
nginx:latest                                                          f9ea18bfa4fa       243MB         66.7MB    U
sonarqube:community                                                   c0f1160bccfa      2.55GB          1.1GB
tomcat:10.1                                                           574ed18cff4d       640MB          174MB    U
ubuntu:24.04                                                          534baea6a22c       119MB         31.7MB
ubuntu:latest                                                         f144425ff09b       162MB         45.6MB    U
```

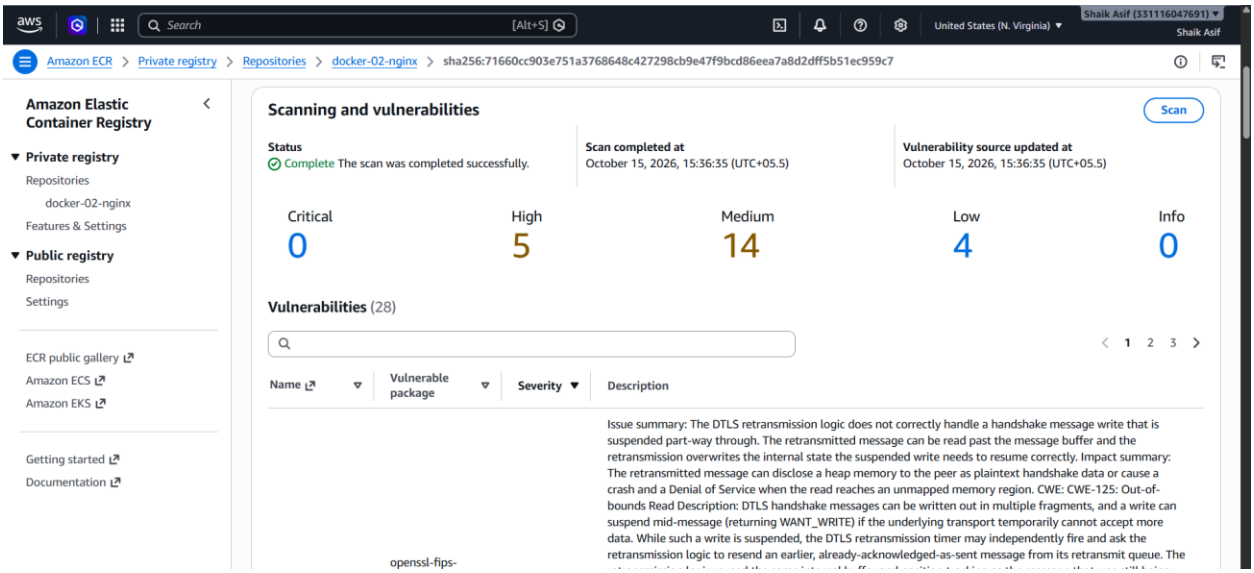
```
ubuntu@ip-172-31-29-148:~$ aws sts get-caller-identity
{
    "UserId": "AROAU2GAN5VFT4ACLLAXZ:i-0c28387df6bacafb1",
    "Account": "331116047691",
    "Arn": "arn:aws:sts::331116047691:assumed-role/EC2-ECR-Push-Role/i-0c28387df6bacafb1"
}
ubuntu@ip-172-31-29-148:~$ aws ecr get-login-password --region us-east-1 | sudo env DOCKER_CONFIG=/home/ubuntu/.docker docker login --username AWS --password-stdin 331116047691.dkr.ecr.us-east-1.amazonaws.com

WARNING! Your credentials are stored unencrypted in '/home/ubuntu/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
```

```
ubuntu@ip-172-31-29-148:~$ sudo env DOCKER_CONFIG=/home/ubuntu/.docker docker push 331116047691.dkr.ecr.us-east-1.amazonaws.com/docker-02-nginx:1.0
The push refers to repository [331116047691.dkr.ecr.us-east-1.amazonaws.com/docker-02-nginx]
6a8f2a3de78b: Pushed
7234d0a9b6ba: Pushed
3803b0a17d94: Pushed
1.0: digest: sha256:71660cc903e751a3768648c427298cb9e47f9bcd86eea7a8d2dff5b51ec959c7 size: 971
ubuntu@ip-172-31-29-148:~$
```





3. Observation

The image scan completed successfully on October 15, 2026.

Parameter	Result
Repository	`docker-02-nginx`
Image tag	`1.0`
Image size	86.76 MB
Scan status	Complete
Vulnerabilities displayed	28
Critical	0
High	5

Medium	14
Low	4
Informational	0

- Vulnerabilities were detected in packages including OpenSSL and Python's pip wheel package.
- Some findings included fixed package versions.
- The severity counts total 23, while the console reports 28 vulnerabilities; these figures do not fully reconcile and should be rechecked in the scan details.

4. Conclusion

Successfully pushed the custom Nginx Docker image to AWS ECR and completed its security scan. The scan identified security vulnerabilities at different severity levels, demonstrating how ECR scanning helps detect risks in container images before deployment.

Task 8: Create a Jenkins Pipeline to Build and Push a Docker Image to Docker Hub

1. Objective

To create a Jenkins declarative pipeline that automatically builds a Docker image, verifies the image, and pushes it to Docker Hub using stored credentials.

2. Steps Used

Step 1: Verified Docker on the EC2 instance

Executed:

```
sudo docker version
```

Verified that Docker Client and Server were installed and running on the EC2 host.

Step 2: Checked the Jenkins container

Executed:

```
sudo docker ps -a
sudo docker images jenkins/jenkins
```

Verified the existing Jenkins image and container environment.

Step 3: Created a Jenkins container

Started Jenkins with port mappings and a persistent volume:

```
sudo docker run -d \
  --name jenkins-lab \
  -p 8080:8080 \
  -p 50000:50000 \
  -v jenkins_home:/var/jenkins_home \
  jenkins/jenkins:lts-jdk21
```

Step 4: Created a custom Jenkins image with Docker CLI

Created a `Dockerfile` inside `~/jenkins-docker`:

```
FROM jenkins/jenkins:lts-jdk21

USER root

RUN apt-get update \
    && apt-get install -y --no-install-recommends docker-cli \
    && rm -rf /var/lib/apt/lists/*

USER jenkins
```

Built the image:

```
sudo docker build --no-cache -t jenkins-docker:lts .
```

Step 5: Recreated Jenkins with Docker access

Stopped and removed the previous Jenkins container while preserving the `jenkins_home` volume.

Started the updated container:

```
sudo docker run -d \
  --name jenkins-lab \
  -p 8080:8080 \
  -p 50000:50000 \
  -v jenkins_home:/var/jenkins_home \
  -v /var/run/docker.sock:/var/run/docker.sock \
  --group-add "$(stat -c '%g' /var/run/docker.sock)" \
  jenkins-docker:lts
```

Verified Docker CLI availability:

```
sudo docker exec jenkins-lab docker --version
```

Verified access to the host Docker engine:

```
sudo docker exec jenkins-lab docker ps
```

Step 6: Created a Jenkins pipeline job

Created a new Jenkins item named `Docker-Hub-Push-Pipeline` and selected Pipeline as the job type.

Initially created a simple pipeline with Build and Test stages and confirmed that it executed successfully.

Step 7: Configured Docker Hub credentials

Added Docker Hub credentials under Jenkins Credentials.

- Kind: Username with password
- Username: Docker Hub username
- Password: Docker Hub access token
- ID: `dockerhub-creds`

Step 8: Configured the declarative pipeline

Added three stages: Build, Test, and Push to Docker Hub.

The pipeline used the following image name:

```
asifshaik6558/docker-02-nginx:jenkins-1
```

The Build stage created an Nginx image with a custom `index.html` welcome page.

The Test stage verified the image using:

```
docker image inspect asifshaik6558/docker-02-nginx:jenkins-1
```

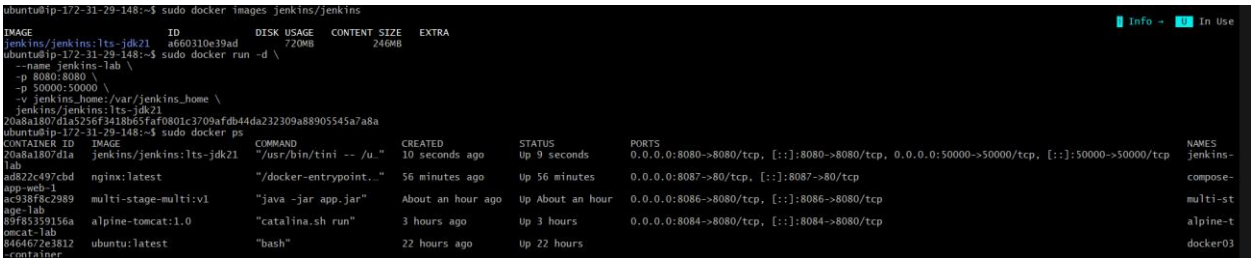
The Push stage authenticated using Jenkins credentials and pushed the image:

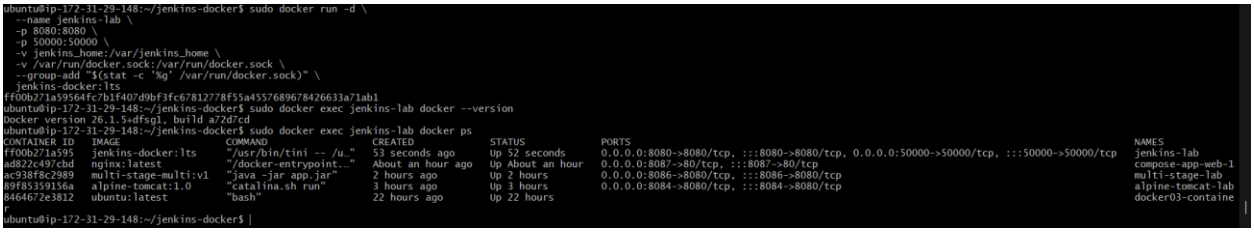
```
docker push asifshaik6558/docker-02-nginx:jenkins-1
```

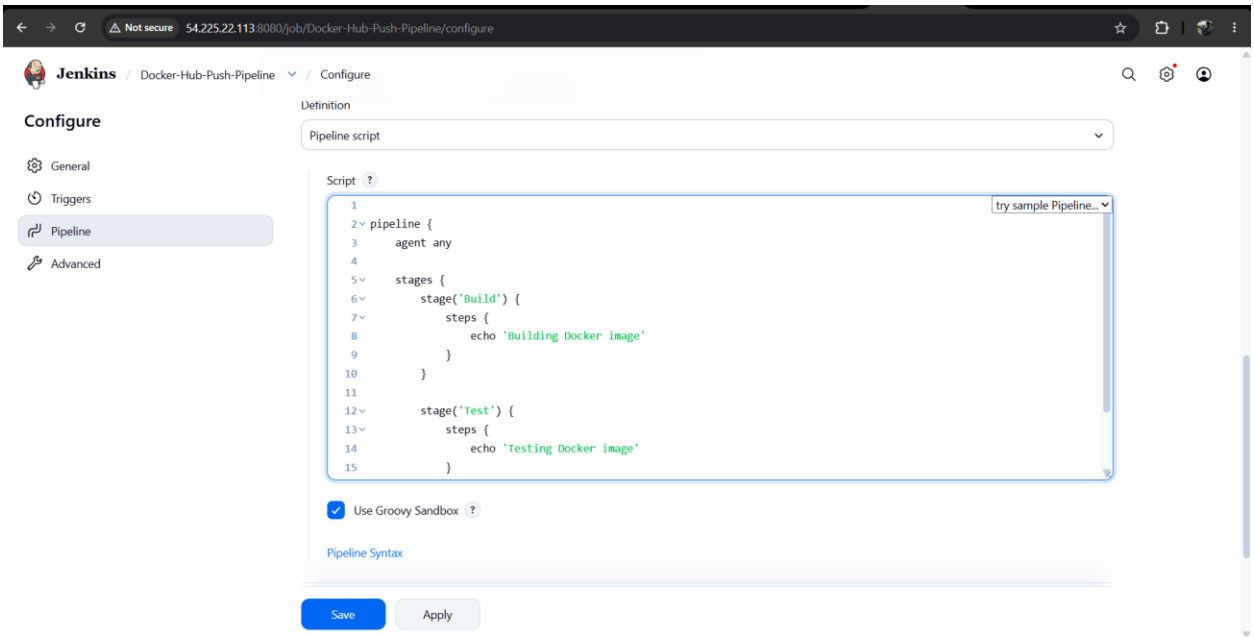
Step 9: Executed the pipeline

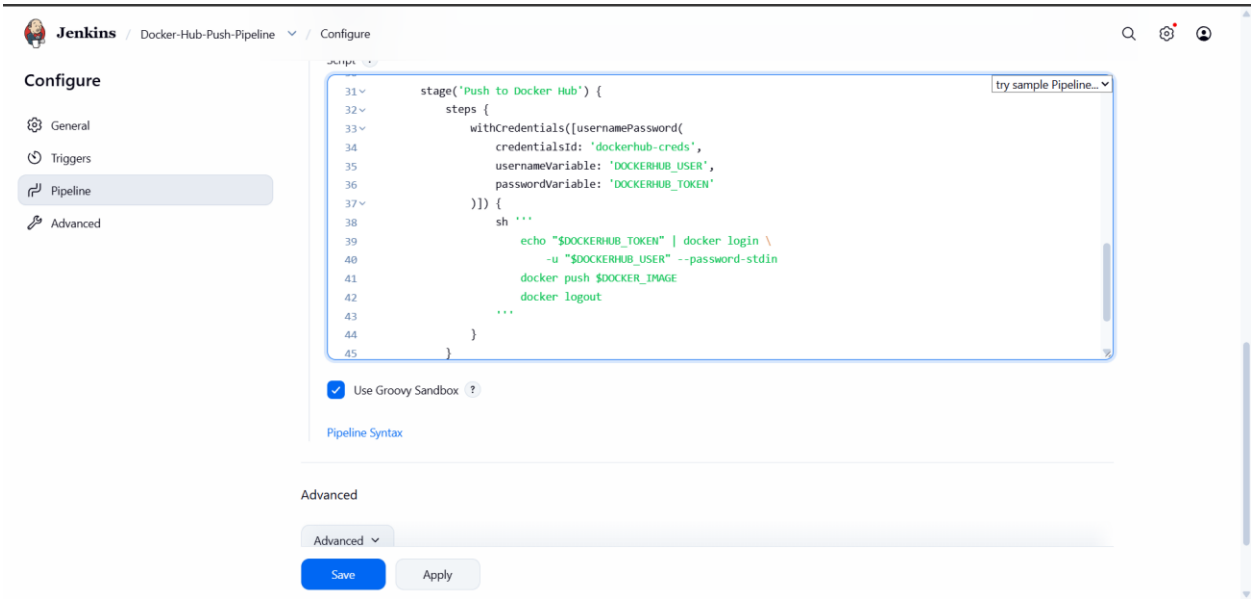
Clicked Build Now and checked the Console Output.

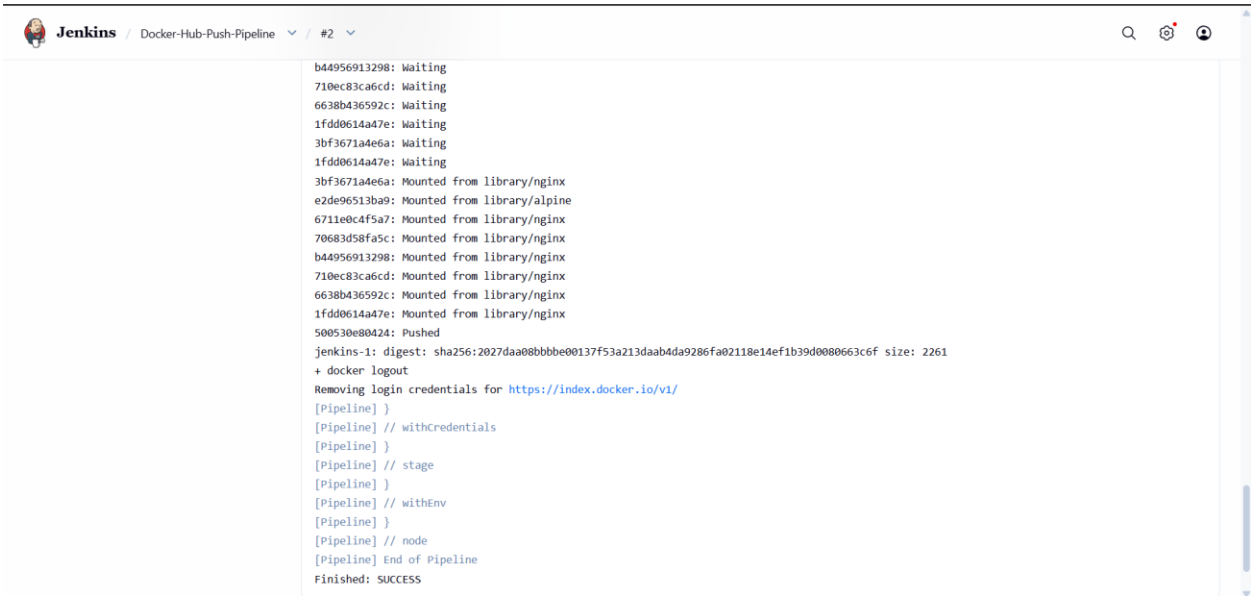
The pipeline completed all three stages successfully.

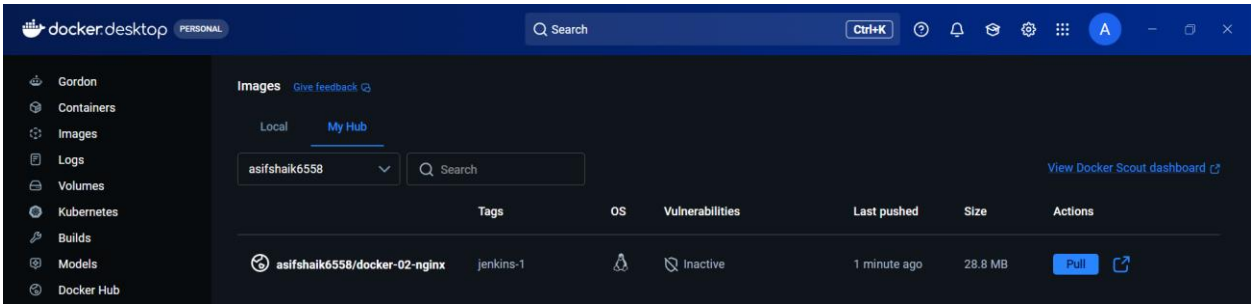





Not secure 54.225.22.113:8080/job/Docker-Hub-Push-Pipeline/configure
Jenkins / Docker-Hub-Push-Pipeline / Configure
Configure
General
Triggers
Pipeline
Advanced
Definition
Pipeline script
Script
try sample Pipeline...
pipeline {
agent any
stages {
stage('Build') {
steps {
echo 'Building Docker image'
}
}
stage('Test') {
steps {
echo 'Testing Docker image'
}
Use Groovy Sandbox
Pipeline Syntax
Save
Apply


Jenkins / Docker-Hub-Push-Pipeline / Configure
Configure
General
Triggers
Pipeline
Advanced
try sample Pipeline...
stage('Push to Docker Hub') {
steps {
withCredentials([usernamePassword(
credentialsId: 'dockerhub-creds',
usernameVariable: 'DOCKERHUB_USER',
passwordVariable: 'DOCKERHUB_TOKEN'
)]) {
sh '''
echo "$DOCKERHUB_TOKEN" | docker login \
-u "$DOCKERHUB_USER" --password-stdin
docker push $DOCKER_IMAGE
docker logout
'''
}
}
Use Groovy Sandbox
Pipeline Syntax
Advanced
Advanced
Save
Apply





3. Observation

- Jenkins was running successfully inside a Docker container.
- Docker CLI was installed in the custom Jenkins image.
- Jenkins accessed the EC2 host's Docker engine through the Docker socket.
- The custom Nginx image was built successfully.
- The image inspection command completed successfully.
- Docker Hub authentication succeeded using Jenkins credentials.
- The image was pushed to Docker Hub with the tag `jenkins-1`.
- The pipeline ended with `Finished: SUCCESS`.

4. Conclusion

Successfully created and executed a Jenkins declarative pipeline to automate Docker image building, image verification, and publishing to Docker Hub. This task demonstrated how Jenkins integrates with Docker and uses stored credentials to automate container image delivery.

Pipeline flow:

Jenkins Pipeline

1. Build Docker Image

2. Test Docker Image

3. Push Image to Docker Hub

SUCCESS